

HDOC - Day 12

Menu-Driven C Program Using Switch-Case

Problem Statement

Create a menu-driven C program using a switch-case statement. The program should accept a positive integer and carry out one of the following tasks according to the option selected by the user:

- Calculate the sum and the product of the first digit and the last digit of the entered number.
- Compare every digit of the number with a given digit k and count how many are smaller than, equal to, or greater than k.

Algorithm

1. Begin the program.
2. Read a positive integer and store it in num.
3. Check whether num is positive. If $\text{num} \leq 0$, print "Please enter a positive integer" and terminate the program.
4. Show a menu containing two options: (1) calculate the sum and product of the first and last digits, and (2) compare the digits of the number with k.
5. Read the option selected by the user and store it in choice.
6. Use `switch(choice)` to execute the selected operation.
7. For case 1, obtain the last digit using $\text{num} \% 10$. Copy num into temp and repeatedly divide temp by 10 until only the first digit remains. Then find $\text{sum} = \text{first} + \text{last}$ and $\text{product} = \text{first} * \text{last}$, and display the first digit, last digit, sum, and product.
8. For case 2, read digit k. If k is not between 0 and 9, display an appropriate message. Otherwise, initialize lesser, equal, and greater to 0. Extract each digit from temp using $\% 10$, compare it with k, update the corresponding counter, and continue after dividing temp by 10. Finally, display all three counts.
9. If the entered menu option does not match any case, use the default case to print "Invalid choice".
10. End the program.

Test Case Table

Test No.	Input Number	Menu Choice	k	Expected Result
1	12345	1	-	First digit = 1, Last digit = 5, Sum = 6, Product = 5
2	12345	2	3	Count_lesser = 2, Count_equal = 1, Count_greater = 2
3	123	3	-	Invalid choice.
4	-12	-	-	Please enter a

				positive integer.
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C Program

```
#include <stdio.h>
```

```
int main()
{
    int num, choice, temp, first, last;
    int sum, product, k, digit;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            last = num % 10;
            temp = num;

            while (temp >= 10)
                temp = temp / 10;

            first = temp;
            printf("First digit = %d\n", first);
            printf("Last digit = %d\n", last);
            sum = first + last;
            product = first * last;

            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("Please enter k between 0 and 9.\n");
                break;
            }

            temp = num;
            while (temp > 0)
            {
                digit = temp % 10;
```

```

        if (digit < k)
            lesser++;
        else if (digit == k)
            equal++;
        else
            greater++;

        temp = temp / 10;
    }

    printf("Count_lesser = %d\n", lesser);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
    break;

default:
    printf("Invalid choice.\n");
}

return 0;
}

```

Compilation and Execution

Compilation command: gcc num.c -o num

Execution command: ./num

Compilation

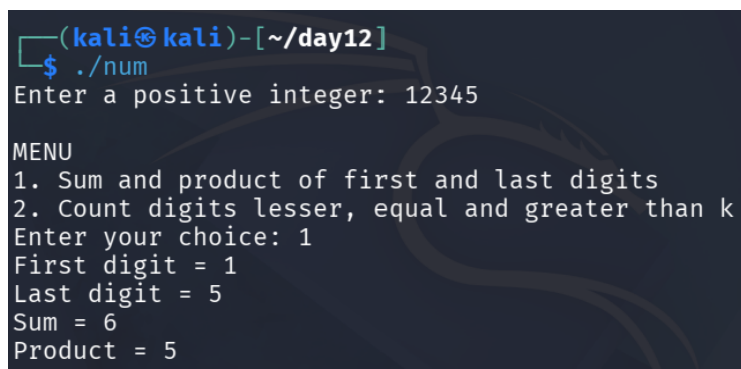


```

(kali㉿kali)-[~/day12]
$ gcc num.c -o num

```

Execution - Choice 1



```

(kali㉿kali)-[~/day12]
$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

```

Execution - Choice 2

```
(kali@kali)-[~/day12]
$ ./num
Enter a positive integer: 12345

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

Execution - Invalid Choice

```
(kali@kali)-[~/day12]
$ ./num
Enter a positive integer: 123

MENU
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 3
Invalid choice.
```

Execution - Invalid Positive Integer

```
(kali@kali)-[~/day12]
$ ./num
Enter a positive integer: -12
Please enter a positive integer.
```